

Lab Assignment — XML: Defining Structure with a DTD

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Aim

Write a Document Type Definition (DTD) that defines the structure of a small library system, then create XML files that conform to it. You will also validate a correct file against your DTD and show what happens when a file breaks the rules.

Scenario

A small library keeps records of its **books**, its **members**, and the **loans** that link a member to a book they have borrowed. You must design the DTD that describes this data and provide XML that matches it.

Tasks

Task 1 Write the DTD [6 marks]

Create a file named `library.dtd` that defines the library structure. Your DTD must include all of the following:

- a root element `library` that contains a set of books, a set of members, and a set of loans
- content models that use the cardinality symbols `?`, `*` and `+` where appropriate
- an attribute of type `ID` that uniquely identifies each book and each member
- an attribute of type `IDREF` used by a loan to link to a book and to a member
- one enumerated attribute that allows only a fixed set of values (for example a book category or a membership type)
- at least one `#REQUIRED` attribute and one `#IMPLIED` attribute
- at least one element whose content is plain text (`#PCDATA`)

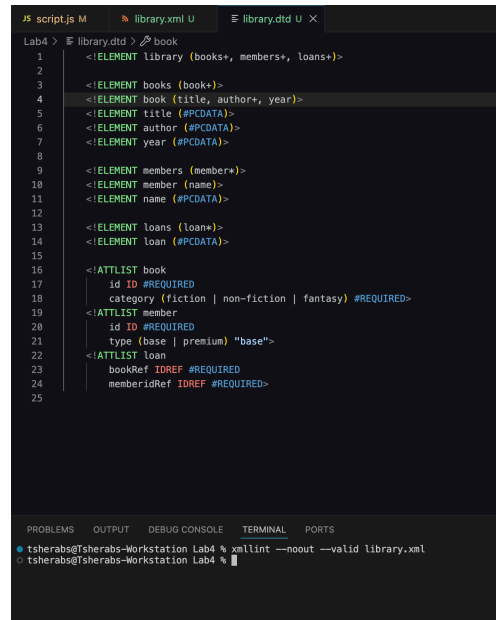
Task 2 Create a valid XML file [5 marks]

Create a file named `library.xml` that references your DTD and conforms to it. It must contain, at minimum:

- at least **four books**, **three members**, and **two loans**
- each loan using `IDREF` values to link one member to one book

Task 3 Validate the file [2 marks]

Validate `library.xml` against `library.dtd` using a tool of your choice (a web browser, the `xmllint` command, or an online DTD validator) and show that it reports the document as **valid**. Paste or screenshot the result.



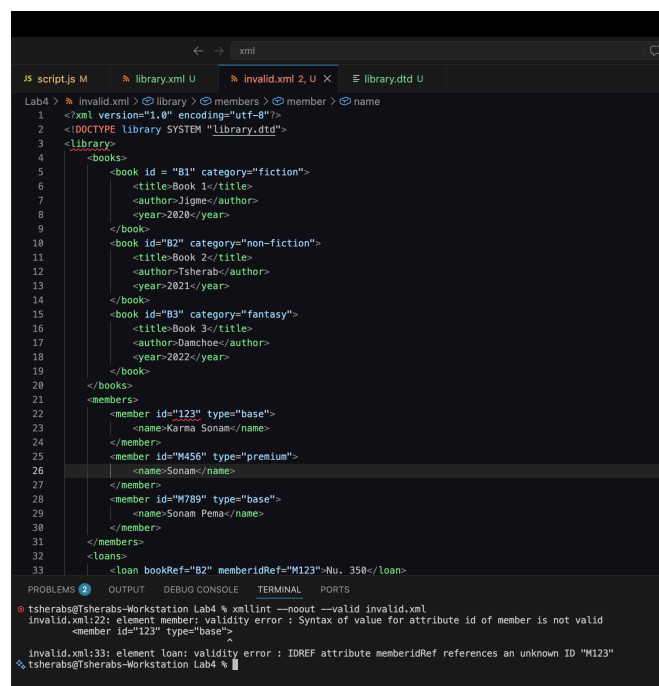
```
1 <!ELEMENT library (books+, members+, loans+)>
2
3 <!ELEMENT books (book+)>
4 <!ELEMENT book (title, author+, year)>
5 <!ELEMENT title (#PCDATA)>
6 <!ELEMENT author (#PCDATA)>
7 <!ELEMENT year (#PCDATA)>
8
9 <!ELEMENT members (member+)>
10 <!ELEMENT member (name)>
11 <!ELEMENT name (#PCDATA)>
12
13 <!ELEMENT loans (loan+)>
14 <!ELEMENT loan (#PCDATA)>
15
16 <!--
17 <!-- id ID #REQUIRED
18 <!-- category (fiction | non-fiction | fantasy) #REQUIRED
19 <!--
20 <!-- id ID #REQUIRED
21 <!-- type (base | premium) "base"
22 <!--
23 <!-- bookRef IDREF #REQUIRED
24 <!-- memberIdRef IDREF #REQUIRED
25
```

```
Lab4 > xmllint --noout --valid library.xml
tsherabs@Tsherabs-Workstation Lab4 %
```

CTE412 · Unit 1 · XML Lab Assignment (DTD)

Task 4 Break a rule [2 marks]

Make a copy named `invalid.xml` and deliberately break **one** DTD rule — for example, remove a `#REQUIRED` attribute or use a value that is not in the enumerated list. Validate it, capture the error message, and state in one line which rule was broken.



```
1 <?xml version="1.0" encoding="utf-8"?>
2 <!DOCTYPE library SYSTEM "library.dtd">
3 <library>
4 <books>
5 <book id="B1" category="fiction">
6 <title>Book 1</title>
7 <author>Jigme</author>
8 <year>2020</year>
9 </book>
10 <book id="B2" category="non-fiction">
11 <title>Book 2</title>
12 <author>Tsherab</author>
13 <year>2021</year>
14 </book>
15 <book id="B3" category="fantasy">
16 <title>Book 3</title>
17 <author>Dandhoe</author>
18 <year>2022</year>
19 </book>
20 </books>
21 <members>
22 <member id="123" type="base">
23 <name>Karma Sonam</name>
24 </member>
25 <member id="W456" type="premium">
26 <name>Sonam</name>
27 </member>
28 <member id="W789" type="base">
29 <name>Sonam Pema</name>
30 </member>
31 </members>
32 <loans>
33 <loan bookRef="B2" memberIdRef="W123">Nu. 350</loan>
34 </loans>
35 </library>
```

```
Lab4 > xmllint --noout --valid invalid.xml
invalid.xml:22: element member: validity error : Syntax of value for attribute id of member is not valid
<member id="123" type="base">
invalid.xml:33: element loan: validity error : IDREF attribute memberIdRef references an unknown ID "W123"
tsherabs@Tsherabs-Workstation Lab4 %
```

What to submit

- `library.dtd`, `library.xml` and `invalid.xml`.
- The validation output for both the valid and the invalid file (paste or screenshots).

Marks

This assignment is worth **15 marks**. It is graded on the correctness of your DTD and XML (Tasks 1–4), evidence of validation, the clarity of your files, and timely submission on the VLE.